class 4th

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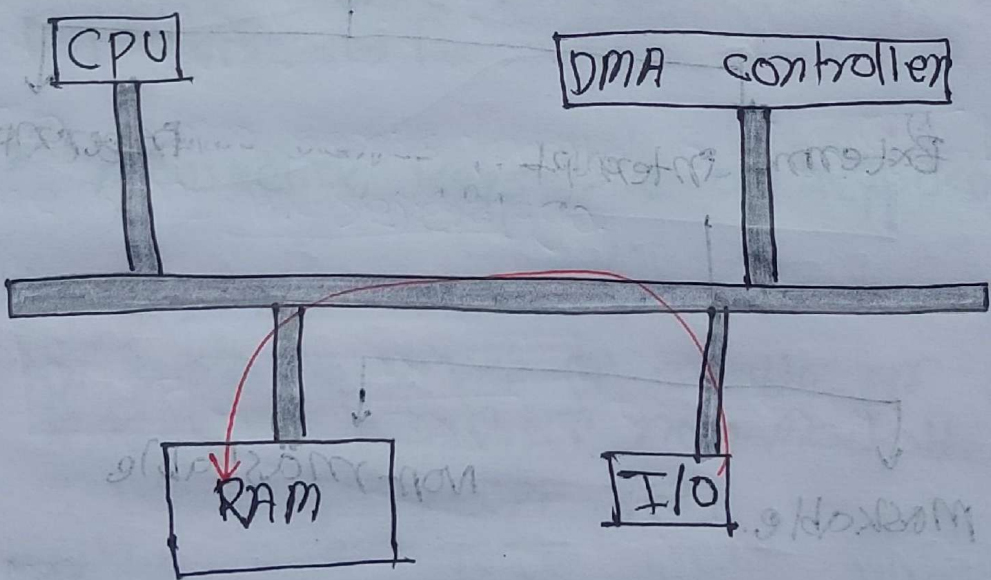
iii) Direct memory access (DMA):

DMA is a type of I/O technique in which data can be transferred between CPU and external device such as the hard disk, without CPU involvement.

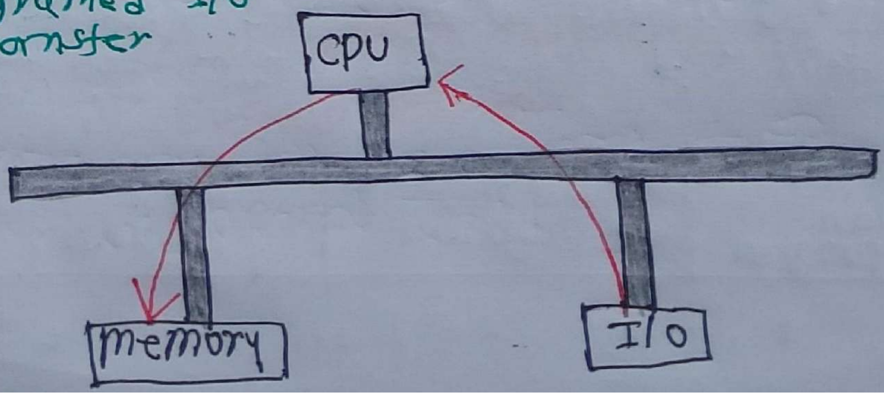
For this work a chip is used called the DMA controller chip 8237

i) DMA transfer

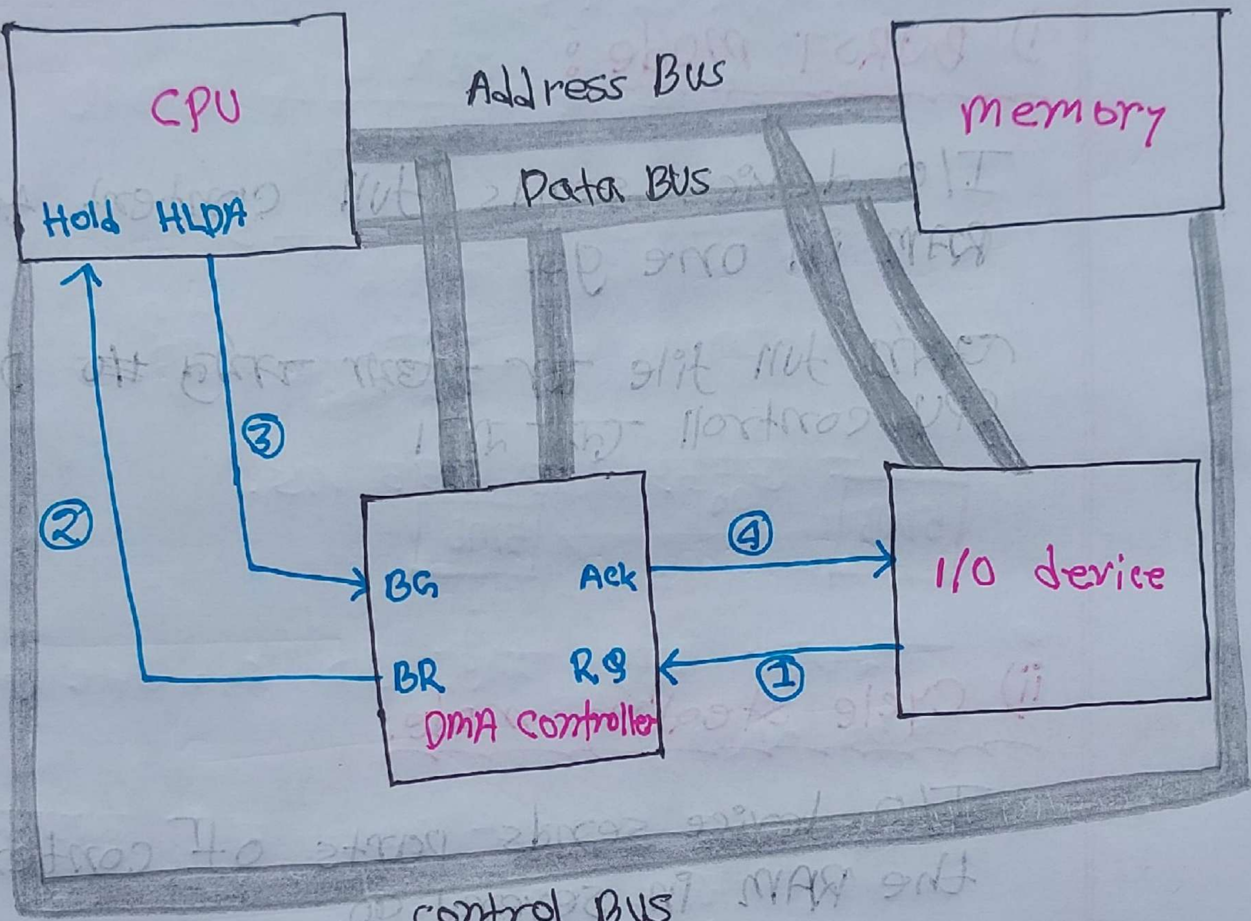


~~DMA transfer~~

ii) programmed I/O transfer



4 steps of DMA :



i) I/O device direct DMA controller এর কাছে Request করে যে or perform করতে চায়।

ii) Bus Request (BR) দিয়ে DMA CPU কে জানায়, Request পেয়ে CPU থেকে বাস controll নেওয়া যায় কিনা।

iii) CPU HLDA Acknowledgement দিয়ে DMA কে BG (Bus grant) pin দিয়ে response দেয়।

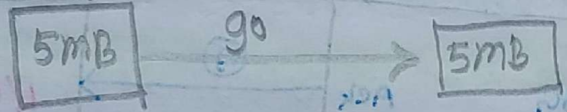
iv) DMA controller I/O device কে জানায় Ack দিয়ে।

DMA 3 types:

i) BURST mode:

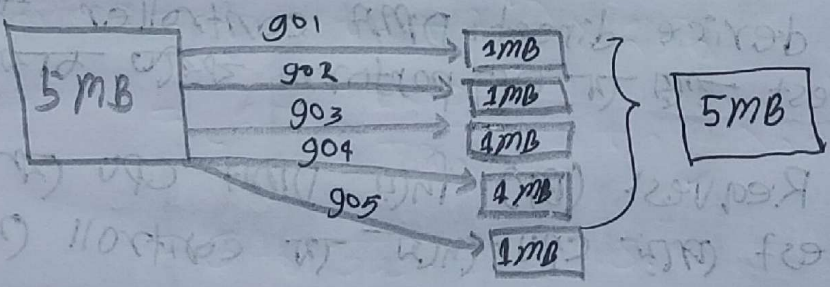
I/O device sends full content to the RAM in one go.

অর্থাৎ full file না যাওয়া পর্যন্ত ~~#o~~ DMA থেকে CPU control নেয় না।



ii) Cycle stealing mode:

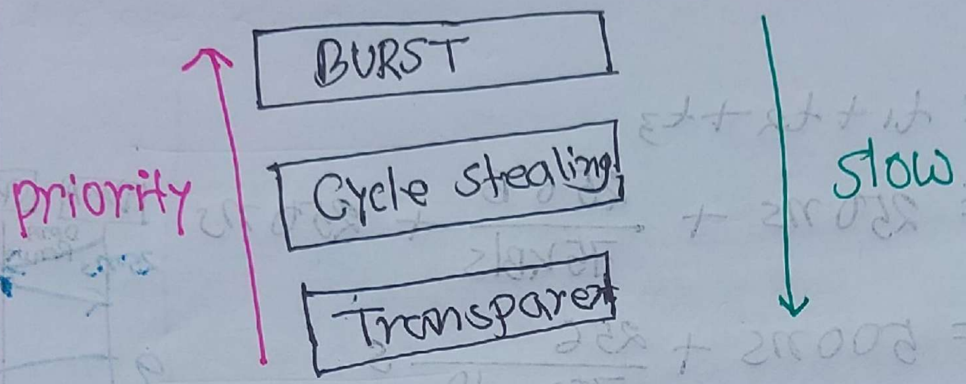
I/O device sends parts of content to the RAM in several go.



N.B প্রতিবারের পাঠানোর সময় Request করা লাগে অর্থাৎ এখানে 5 বার Request করা লাগছে।

iii) Transparent mode:

যখন CPU full free থাকে তখনই CPU DMA কে control দেয়।



Ex:

Transfer of Bus control in either direction, from CPU → device, or device → CPU, take 250 ns. One of the I/O device has data transfer rate of 75 KB/s and employs DMA.

a) Suppose we employs DMA in a BURST mode. How long does it takes to transfer a Block of 256 Bytes?

b) Data are transferred one byte at a time. Calculate the same for Cycle stealing mode.

N.B

$$time = (t_1 + t_2 + t_3)$$

t_1: CPU to device time for Request ~~cent~~ cancel

t_2: Time to transfer Data (content)

t_3: device to cpu time for ~~centre~~ making Request a)

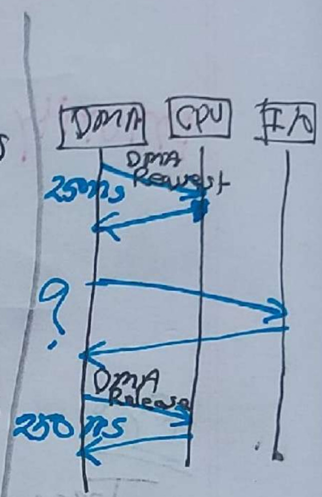
$$\text{time} = t_1 + t_2 + t_3$$

$$= 250\,ns + \frac{256\,B}{75\,KB/s} + 250\,ns$$

$$= 500\,ns + \frac{256}{75 \times 2^{10}}\,s$$

$$= 500\,ns + \frac{256}{75 \times 2^{10}} \times 10^{9}\,ns$$

$$= 3333583.333\,ns.$$



b) ~~একবারে~~ 1 Byte যাবে, so 256 Byte যাবে 256 বারে।

256 বার Request / Cancel করা লাগবে।

$$\text{time} = (\underbrace{t_1 + t_2 + t_3}_{\text{time for 1 Byte}}) \times 256$$

$$= \left(250 + \frac{1 \times 10^{9}}{75 \times 2^{10}}\,ns + 250\right) \times 256$$

$$= 3461333.333\,ns.$$

$1\,KB \rightarrow 2^{10}\,B$
$1\,MB \rightarrow 2^{20}\,B$
$1\,GB \rightarrow 2^{30}\,B$
$1\,TB \rightarrow 2^{40}\,B$

$1\,s \rightarrow 10^{9}\,ns$
$1\,s \rightarrow$

Summer 25 Q 1(a)

CPU $\longrightarrow$ device : 200 ns

device $\longrightarrow$ CPU : 300 ns

time : 148, ~~5~~456,500 ns

file : 3648 Byte

Transfer rate : 1440 kB/m

$$= \frac{1440 \times 2^{10}}{60} \text{ B/s}$$

$$= \frac{1440 \times 10^{-9} \times 2^{10}}{60} \text{ B/ns}$$

Mainly এখানে ব্যবহারে transfer হয়েছে তাই আগের থের ব্যালেন্স হবে।

$$\text{Time} : (t_1 + t_2 + t_3) \times \frac{3648}{n}$$ → n Byte one transfer at a time

$$\Rightarrow 148456500 = \left(200\,ns + \frac{n \times 10^{9}\,ns}{\frac{1440}{60} \times 2^{10}} + 300\,ns\right) \times \frac{3648}{n}$$

$$\Rightarrow 148456500\,n = 729600 + 148437500\,n + 1094400$$

$$\Rightarrow 19000\,n = 1824000$$

$$\Rightarrow n = 96 \text{ Byte}$$